photons each second. In the quantum world, we've learned that everything has both particlelike and wavelike attributes.

Over the last eight decades, the ubiquity and utility of quantum mechanical probability waves to predict and explain experimental results has been established beyond any doubt. Yet there is still no universally agreed-upon way to envision what quantum mechanical probability waves actually are. Whether we should say that an electron's probability wave *is* the electron, or that it's *associated* with the electron, or that it's a *mathematical device* for describing the electron's motion, or that it's the *embodiment of what we can know* about the electron is still debated. What is clear, though, is that through these waves, quantum mechanics injects probability into the laws of physics in a manner that no one had anticipated. Meteorologists use probability to predict the likelihood of rain. Casinos use probability to predict the likelihood you'll throw snake eyes. But probability plays a role in these examples because we haven't all of the information necessary to make definitive predictions. According to Newton, if we knew in complete detail the state of the environment (the positions and velocities of every one of its particulate ingredients), we would be able to predict (given sufficient calculational prowess) with certainty whether it will rain at 4:07 p.m. tomorrow; if we knew all the physical details of relevance to a craps game (the precise shape and composition of the dice, their speed and orientation as they left your hand, the composition of the table and its surface, and so on), we would be able to predict with certainty how the dice will land. Since, in practice, we can't gather all this information (and, even if we could, we do not yet have sufficiently powerful computers to perform the calculations required to make such predictions), we set our sights lower and predict only the probability of a given outcome in the weather or at the casino, making reasonable guesses about the data we don't have.

The probability introduced by quantum mechanics is of a different, more fundamental character. Regardless of improvements in data collection or in computer power, the best we can ever do, according to quantum mechanics, is predict the probability of this or that outcome. The best we can ever do is predict the probability that an electron, or a proton, or a neutron, or any other of nature's constituents, will be found here or there. Probability reigns supreme in the microcosmos.